

Lab 2: Basic Pandas Operations

Objective:

In this lab, you will learn how to import a CSV file into a pandas DataFrame and perform basic operations such as viewing the first few rows, the last few rows, and generating summary statistics of the DataFrame.

Step-by-Step Instructions:

1. Import the pandas Library

First, we need to import the pandas library. Pandas is a powerful library for data manipulation and analysis in Python.

In [1]: `import pandas as pd`

2. Load a CSV File into a DataFrame

Let's load a sample CSV file into a DataFrame. For this demo, we'll use a CSV file named `students_scores.csv` . Make sure this file is in your working directory or provide the full path to the file.

In [3]: `df = pd.read_csv('students_scores.csv')`

3. View the First Few Rows of the DataFrame

To get a quick look at the first few rows of the DataFrame, we use the `head()` method. By default, it shows the first 5 rows, but you can specify the number of rows to display.

In [5]: `print("First 5 rows of the DataFrame:")
print(df.head())`

First 5 rows of the DataFrame:

	Name	Math	Science	English
0	Alice	85	92	88
1	Bob	78	83	80
2	Charlie	90	95	85
3	David	70	75	72
4	Eve	88	90	92

4. View the Last Few Rows of the DataFrame

Similarly, to see the last few rows, we use the `tail()` method. It also shows 5 rows by default, but you can change this number as well.

In [7]: `print("Last 5 rows of the DataFrame:")
print(df.tail())`

Last 5 rows of the DataFrame:

	Name	Math	Science	English
0	Alice	85	92	88
1	Bob	78	83	80
2	Charlie	90	95	85
3	David	70	75	72
4	Eve	88	90	92

5. Generate Summary Statistics

The `describe()` method provides a quick overview of the numerical columns in the DataFrame, including count, mean, standard deviation, min, and max values, as well as the 25th, 50th (median), and 75th percentiles.

In [9]: `print("Summary statistics of the DataFrame:")
print(df.describe())`

Summary statistics of the DataFrame:

	Math	Science	English
count	5.000000	5.000000	5.000000
mean	82.200000	87.000000	83.400000
std	8.197561	8.031189	7.733046
min	70.000000	75.000000	72.000000
25%	78.000000	83.000000	80.000000
50%	85.000000	90.000000	85.000000
75%	88.000000	92.000000	88.000000
max	90.000000	95.000000	92.000000

Exercise:

1. Create your own CSV file with different data (e.g., sales data, employee data).
2. Load your CSV file into a DataFrame.
3. Display the first 10 rows of your DataFrame.
4. Display the last 3 rows of your DataFrame.
5. Generate summary statistics for your DataFrame.

Feel free to ask questions if you have any trouble or need further assistance. Happy coding!

In [19]: `#1/2
df = pd.read_csv("avocado.csv")
#3
df.head(10)`

Out[19]:	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62	93.25	0.0	conventional	2015	Albany
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07	97.49	0.0	conventional	2015	Albany
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21	103.14	0.0	conventional	2015	Albany
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40	133.76	0.0	conventional	2015	Albany
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26	197.69	0.0	conventional	2015	Albany
5	5	2015-11-22	1.26	55979.78	1184.27	48067.99	43.61	6683.91	6556.47	127.44	0.0	conventional	2015	Albany
6	6	2015-11-15	0.99	83453.76	1368.92	73672.72	93.26	8318.86	8196.81	122.05	0.0	conventional	2015	Albany
7	7	2015-11-08	0.98	109428.33	703.75	101815.36	80.00	6829.22	6266.85	562.37	0.0	conventional	2015	Albany
8	8	2015-11-01	1.02	99811.42	1022.15	87315.57	85.34	11388.36	11104.53	283.83	0.0	conventional	2015	Albany
9	9	2015-10-25	1.07	74338.76	842.40	64757.44	113.00	8625.92	8061.47	564.45	0.0	conventional	2015	Albany

In [21]: `#4
df.tail(3)`

Out[21]:	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	type	year	region
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9351.80	42.31	0.0	organic	2018	WestTexNewMexico
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919.54	50.00	0.0	organic	2018	WestTexNewMexico
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11988.14	26.01	0.0	organic	2018	WestTexNewMexico

In [23]: `#5
df.describe()`

Out [23]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags	year
count	18249.000000	18249.000000	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04	18249.000000	18249.000000
mean	24.232232	1.405978	8.506440e+05	2.930084e+05	2.951546e+05	2.283974e+04	2.396392e+05	1.821947e+05	5.433809e+04	3106.426507	2016.147899
std	15.481045	0.402677	3.453545e+06	1.264989e+06	1.204120e+06	1.074641e+05	9.862424e+05	7.461785e+05	2.439660e+05	17692.894652	0.939938
min	0.000000	0.440000	8.456000e+01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000	2015.000000
25%	10.000000	1.100000	1.083858e+04	8.540700e+02	3.008780e+03	0.000000e+00	5.088640e+03	2.849420e+03	1.274700e+02	0.000000	2015.000000
50%	24.000000	1.370000	1.073768e+05	8.645300e+03	2.906102e+04	1.849900e+02	3.974383e+04	2.636282e+04	2.647710e+03	0.000000	2016.000000
75%	38.000000	1.660000	4.329623e+05	1.110202e+05	1.502069e+05	6.243420e+03	1.107834e+05	8.333767e+04	2.202925e+04	132.500000	2017.000000
max	52.000000	3.250000	6.250565e+07	2.274362e+07	2.047057e+07	2.546439e+06	1.937313e+07	1.338459e+07	5.719097e+06	551693.650000	2018.000000

In []: